| Ex.No.11 | STUDY OF SOCKET PROGAMMING IN PYTHON |
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**Aim:**

To study the concepts of socket programming in python.

**STUDY OF SOCKET PROGAMMING IN PYTHON**

A network socket is one endpoint of a two-way communication link between two programs running on a network. It is a software-based handle combining an IP address and a port number, allowing different processes to send and receive data across a device or the internet

**Socket programming in python**

- Python's core networking library is Socket Module.
- Python's socket module has both class-based and instances-based methods.
- Class-based method is an intuitive approach which doesn't need an instance of a
- socket object.
- For example, in order to print a machine's IP address, you don't need a socket
- object. Instead, just call the socket's class-based methods.
- In instance-based method, if some data needed to be sent to a server application,
- it is more intuitive to create a socket object to perform that explicit operation.
- This module has everything you need to build socket servers and clients.

**Algorithm 1: TCP Server**

**Input:** Client message
**Output:** Response message to the client

1. **Start**
2. Import the socket module.

3. Create a TCP socket using socket().
4. Bind the socket to the IP address localhost and port number 5000.
5. Put the server socket into listening mode using listen().
6. Wait for a client connection using accept().
7. Receive the message from the client using recv().
8. Decode the received message.
9. Display the client's message.
10. Create a response message.
11. Encode and send the response to the client using send().
12. Close the client connection.
13. Close the server socket.
14. **Stop**

**TCP Client**

**Input:** Message to be sent to the server
**Output:** Response received from the server

1. **Start**
2. Import the socket module.
3. Create a TCP socket using socket().
4. Connect to the server using connect() with localhost and port 5000.
5. Create a message to send to the server.
6. Encode the message.
7. Send the message to the server using send().
8. Wait for the server's response using recv().
9. Decode the received response.
10. Display the server's response.
11. Close the client socket.
12. **Stop**

**PROGRAM(Server):**

```
# TCP Server Program
import socket
# Create a socket
server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
# Bind IP address and port
server_socket.bind(("localhost", 5000))
# Listen for client connection
server_socket.listen(1)
print("Server is waiting for connection...")
# Accept client connection
conn, address = server_socket.accept()
print("Client connected:", address)
# Receive message from client
message = conn.recv(1024).decode()
print("Message from client:", message)
# Send response to client
response = "Hello Client! Message received successfully."
conn.send(response.encode())
# Close connection
conn.close()
server_socket.close()
```

**PROGRAM (CLIENT):**

```
# TCP Client Program
import socket
# Create a socket
```

```
client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
# Connect to server
client_socket.connect(("localhost", 5000))
# Send message to server
message = "Hello Server!"
client_socket.send(message.encode())
# Receive response from server
response = client_socket.recv(1024).decode()
print("Message from server:", response)
# Close connection
client_socket.close()
```

**SAMPLE OUTPUT:**

**Server Output**

```
C:\Python> python server.py

Server is waiting for connection...
Client connected: ('127.0.0.1', 54321)
Message from client: Hello Server!
```

**Client Output**

```
C:\Python> python client.py

Message from server: Hello Client! Message received successfully.
```

**SERVER OUTPUT:**

```
PS C:\WINDOWS\system32> cd "C:\Users\REC\AppData\Local\Programs\Python\Python314"
PS C:\Users\REC\AppData\Local\Programs\Python\Python314> & "C:\Users\REC\AppData\Local\Programs\P
ython\Python314\python.exe" "C:\Users\REC\AppData\Local\Programs\Python\Python314\exp10s.py"
Server is waiting for connection...
Client connected: ('127.0.0.1', 50506)
Message from client: Hello Server!
PS C:\Users\REC\AppData\Local\Programs\Python\Python314>
```

**CLIENT OUTPUT:**

```
PS C:\WINDOWS\system32> cd "C:\Users\REC\AppData\Local\Programs\Python\Python314"
PS C:\Users\REC\AppData\Local\Programs\Python\Python314> & "C:\Users\REC\AppData\Local\Programs\P
ython\Python314\python.exe" "C:\Users\REC\AppData\Local\Programs\Python\Python314\exp10c.py"
Message from server: Hello Client! Message received successfully
PS C:\Users\REC\AppData\Local\Programs\Python\Python314>
```

**RESULT:**

Thus the experiment to study the socket programming in python has been done and the output is verified.